Операционные системы

Продвинутое использование git

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2025-12-22

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1. Цели и задачи работы

Цель лабораторной работы

Получение навыков правильной работы с репозиториями git.

Задачи лабораторной работы

1 Выполнить работу для тестового репозитория.

2 Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

2. Процесс выполнения лабораторной работы

Node.js

```
aaamrahov@aaamrahov:~$
aaamrahov@aaamrahov:~$ pnpm setup
Appended new lines to /home/aaamrahov/.bashrc

Next configuration changes were made:
export PNPM_HOME="/home/aaamrahov/.local/share/pnpm"
case ":$PATH:" in
  *":$PNPM_HOME:"*) ;;
  *) export PATH="$PNPM_HOME:$PATH" ;;
esac

To start using pnpm, run:
source /home/aaamrahov/.bashrc
aaamrahov@aaamrahov:~$ source ~/.bashrc
aaamrahov@aaamrahov:~$
```

Рисунок 1: Node.js

commitizen

```
aaamrahov@aaamrahov:~$ pnpm add -g commitizen

    Update available! 10.27.0 → 10.30.0.
    Changelog: https://pnpm.io/v/10.30.0
      To update, run: pnpm self-update

 WARN  2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +149
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
Progress: resolved 149, reused 0, downloaded 149, added 149, done

/home/aaamrahov/.local/share/pnpm/global/5:
+ commitizen 4.3.1

Done in 2.5s using pnpm v10.27.0
aaamrahov@aaamrahov:~$
```

Рисунок 2: установка commitizen

standard-changelog

```
aaamrahov@aaamrahov:~$ pnpm add -g standard-changelog
 WARN  2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +32
++++++++++++++++++++++++++++++++
Progress: resolved 181, reused 149, downloaded 32, added 32, done

/home/aaamrahov/.local/share/pnpm/global/5:
+ standard-changelog 7.0.1

Done in 2.7s using pnpm v10.27.0
aaamrahov@aaamrahov:~$
```

Рисунок 3: установка standard-changelog

package.json

Open

package.json
~/git-extended

```
1  {
2      "name": "git-extended",
3      "version": "1.0.0",
4      "description": "Git repo for educational purposes",
5      "main": "index.js",
6      "repository": "git@github.com:AmrakhovAzer/git-extended.git",
7      "author": "AmrakhovAzer",
8      "license": "CC-BY-4.0",
9      "config": {
10         "commitizen": {
11             "path": "cz-conventional-changelog"
12         }
13     }
14 }
```

Рисунок 4: package.json

Использование commitizen

```
aaamrahov@aaamrahov:~/git-extended$
aaamrahov@aaamrahov:~/git-extended$ git add .
aaamrahov@aaamrahov:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:    Other changes that don't modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
 (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master bf0f264] chore: test
 1 file changed, 14 insertions(+)
 create mode 100644 package.json
aaamrahov@aaamrahov:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.12 KiB | 1.12 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:AmrakhovAzer/git-extended.git
   8d5d472..bf0f264  master -> master
aaamrahov@aaamrahov:~/git-extended$
```

Рисунок 5: Отправка

```
aaamrahov@aaamrahov:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
   - master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/aaamrahov/git-extended/.git/hooks]
aaamrahov@aaamrahov:~/git-extended$ git branch
* develop
  master
aaamrahov@aaamrahov:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:      https://github.com/AmrakhovAzer/git-extended/pull/new/develop
remote:
To github.com:AmrakhovAzer/git-extended.git
 * [new branch]      develop -> develop
aaamrahov@aaamrahov:~/git-extended$
```

Рисунок 6: Инициализация и загрузка изменений

git-flow

```
Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

     git flow release finish '1.0.0'

aaamrahov@aaamrahov:~/git-extended$ standard-changelog --first-release
aaamrahov@aaamrahov:~/git-extended$ git add CHANGELOG.md
aaamrahov@aaamrahov:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 ae7fcf7] chore(site): add changelog
 1 file changed, 4 insertions(+)
 create mode 100644 CHANGELOG.md
aaamrahov@aaamrahov:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 4 ++++
 1 file changed, 4 insertions(+)
 create mode 100644 CHANGELOG.md
Already on 'master'
Your branch is ahead of 'origin/master' by 2 commits.
  (use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 4 ++++
 1 file changed, 4 insertions(+)
 create mode 100644 CHANGELOG.md
```

Рисунок 7: Завершение релиза

Отправим данные на github

```
aaamrahov@aaamrahov:~/git-extended$
aaamrahov@aaamrahov:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 2.67 KiB | 2.67 MiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:AmrakhovAzer/git-extended.git
   bf0f264..74ecc7d  develop -> develop
   bf0f264..cbeb9e7  master -> master
aaamrahov@aaamrahov:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 159 bytes | 159.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:AmrakhovAzer/git-extended.git
 * [new tag]         v1.0.0 -> v1.0.0
aaamrahov@aaamrahov:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/AmrakhovAzer/git-extended/releases/tag/v1.0.0
aaamrahov@aaamrahov:~/git-extended$
```

Рисунок 8: Отправка

Объединение веток

```
aaamrahov@aaamrahov:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

     git flow feature finish feature_branch

aaamrahov@aaamrahov:~/git-extended$ touch test2
aaamrahov@aaamrahov:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was 74ecc7d).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

aaamrahov@aaamrahov:~/git-extended$
```

Рисунок 9: Объединение веток

```
Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged 'v1.2.3'
- Release tag 'v1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

aaamrahov@aaamrahov:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 2.79 KiB | 2.79 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:AmrakhovAzer/git-extended.git
   74ecc7d..e590fbc  develop -> develop
   cbeb9e7..4468e33  master -> master
aaamrahov@aaamrahov:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 159 bytes | 159.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:AmrakhovAzer/git-extended.git
 * [new tag]         v1.2.3 -> v1.2.3
aaamrahov@aaamrahov:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/AmrakhovAzer/git-extended/releases/tag/v1.2.3
aaamrahov@aaamrahov:~/git-extended$
```

Рисунок 10: Завершение релиза

Подготовка рабочего репозитория

```
aaamrahov@aaamrahov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git add .
aaamrahov@aaamrahov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:    Other changes that don't modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
 (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master a5f939a] chore: test
 1 file changed, 11 insertions(+), 11 deletions(-)
aaamrahov@aaamrahov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.11 KiB | 1.11 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:AmrakhovAzer/2026-1--study--os-intro.git
   076c427..a5f939a  master -> master
aaamrahov@aaamrahov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 11: package.json и коммит

Подготовка рабочего репозитория

```
Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged 'v1.0.0'
- Release tag 'v1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

aaamrahov@aaamrahov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 2.76 KiB | 2.76 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:AmrakhovAzer/2026-1--study--os-intro.git
   a5f939a..5be6ed4  develop -> develop
   a5f939a..a153e73  master -> master
aaamrahov@aaamrahov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 158 bytes | 158.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:AmrakhovAzer/2026-1--study--os-intro.git
 * [new tag]         v1.0.0 -> v1.0.0
aaamrahov@aaamrahov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 12: Завершение релиза

3. Выводы по проделанной работе

Вывод

Мы приобрели практические навыки взаимодействия с дополнительными функциями гитхаб.